EXPERIMENT 11 - PYTHON CODE

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# EXPERIMENT 11 - Family Tree
parents = [
    ("Ravi", "Arun"),
    ("Ravi", "Meena"),
    ("Arun", "Kiran"),
    ("Meena", "Priya"),
    ("Suresh", "Ravi")
]

def is_parent(parent, child):
    return (parent, child) in parents

def get_children(parent):
    return [child for p, child in parents if p == parent]

def get_parents(child):
    return [parent for parent, c in parents if c == child]

def get_grandparents(person):
    grandparents = []
    for parent in get_parents(person):
        grandparents.extend(get_parents(parent))
    return grandparents

person = "Kiran"
print("FAMILY TREE RELATIONSHIPS")
print("=" * 35)
print("\nStep 1: Family Facts")
for parent, child in parents:
    print(f"  {parent} is a parent of {child}")

print("\nStep 2: Find Parents")
print("  Person:", person)
print("  Parents of", person, ":", get_parents(person))

print("\nStep 3: Find Children")
print("  Children of Ravi:", get_children("Ravi"))

print("\nStep 4: Find Grandparents")
for parent in get_parents(person):
    print(f"  Parents of {parent}:", get_parents(parent))
print("  Grandparents of", person, ":", get_grandparents(person))

print("\nStep 5: Check Relationships")
print("  Is Ravi a parent of Arun?", is_parent("Ravi", "Arun"))
print("  Is Suresh a grandparent of Kiran?", "Suresh" in get_grandparents("Kiran"))

print("\n--- Final Result ---")
print("Parents of Kiran:", get_parents("Kiran"))
print("Children of Ravi:", get_children("Ravi"))
print("Grandparents of Kiran:", get_grandparents("Kiran"))
print("Family relationships identified successfully.")
```